UniGuide AI: A RAG-Based Assistant for Answering Admissions Questions

University of New Haven

NLP Project

Professor Khaled Sayed

April 24, 2026

Presented By: Faryal Bahawi



Made with GAMMA

PROJECT OVERVIEW

Project Topic and Domain

1

Project Topic

- Development of a Retrieval-Augmented Generation (RAG) system for admissions question answering
- Retrieves relevant admissions information and generates accurate, grounded responses

2

Domain

- Higher Education (University Admissions)
- Focused on admissions at the University of New Haven

3

Corpus

- University of New Haven admissions website pages
- Application requirements and admissions guidelines
- Application deadlines and important dates
- Admissions FAQ pages
- Selected academic program pages (for basic program information)

Made with GAMMA

OBJECTIVES

Project Objectives

Retrieval-Grounded Answers

Retrieve relevant information from official University of New Haven admissions sources instead of generating unsupported responses

Accurate and Context-Aware Responses

Generate precise and grounded answers based on retrieved admissions content

Reduced Hallucination

Minimize incorrect or fabricated information compared to standard language models

Clear Student-Focused Support

Provide clear and concise answers to help prospective students understand the admissions process

Made with GAMMA



Why This Project Matters

The problem isn't missing information; it's **difficult access and fragmented interpretation**. Students waste time navigating dozens of departmental pages only to contact the wrong office.

Fragmented Web Presence

Information is scattered across multiple pages, departments, and formats with no unified entry point

Misdirected Inquiries

Students frequently contact the wrong office for straightforward questions, creating delays on both sides

Repetitive Admin Load

Staff spend disproportionate time answering the same common questions that could be automated reliably

This project addresses these challenges by providing a RAG-based system that delivers accurate, centralized, and reliable admissions information.

Made with GAMMA

ARCHITECTURE

Proposed System Architecture

1. **Data Collection:** 7 UNH admissions .txt files (deadlines, requirements, tuition, FAQs)
2. **Text Chunking:** Split into ≤300 char chunks, respecting section boundaries
3. **Embedding:** all-MiniLM-L6-v2, 384-dim semantic vectors
4. **Retrieval:** Cosine similarity + keyword boosting, top_k=5 chunks per query
5. **Generation:** Prompt + context → LLM answer, Gemma-2b-it & DistilGPT2

Made with GAMMA

METHODOLOGY

Evaluation Methodology

- **Evaluation Setup**
  A set of domain-specific admissions questions will be used to test the system
- **Evaluation Metrics**
  Responses will be evaluated using **manual assessment based on defined criteria**, including:
  - Accuracy
  - Relevance
  - Groundedness
  - Clarity
- **Model Comparison**
  Outputs from multiple LLMs will be compared across these metrics
- **RAG Effectiveness**
  Responses generated with RAG will be compared to standard LLM outputs to assess improvements in accuracy and reliability

Made with GAMMA

Evaluation Results

3.7/5	1.05/5	10/10
Gemma Average Score	DistilGPT2 Average Score	Questions Gamma Won

Question	Gemma	DistilGPT2	Winner
Priority deadline for intl. grad students?	5.0	1.0	**Gemma**
Priority deadline for intl. grad students?	4.5	1.0	**Gemma**
GRE/GMAT requirement for grad applicants?	4.0	1.0	**Gemma**
English test scores for intl. students?	4.0	1.0	**Gemma**
Docs for international grad applicants?	3.5	1.0	**Gemma**
Docs for transfer undergrad applicants?	4.0	1.0	**Gemma**
Requirements for first-year undergrad?	5.0	1.0	**Gemma**
Total cost of attendance on campus?	3.0	1.0	**Gemma**

Made with GAMMA

Key Findings

Instruction Tuning is Critical

Gemma-2b-it consistently followed the one-sentence format and stayed grounded in context. DistilGPT2 looped, repeated the question, or hallucinated on every question, proving that retrieval alone is not enough without an instruction-tuned generator.

RAG Reduces Hallucination

With retrieved context, Gemma produced factually correct answers for 8 out of 10 questions. Without context, neither model could answer domain-specific UNH questions at all.

Retrieval Quality Matters

> Q3, Q6, and Q8 initially failed because the retriever pulled undergrad/homeschool chunks for graduate queries. Adding source-based boosting and undergrad penalties fixed this , showing that retriever engineering is as important as the generator.

Q3, Q6, and Q8 initially failed because the retriever pulled undergrad/homeschool chunks for graduate queries. Adding source-based boosting and undergrad penalties fixed this — showing that retriever engineering is as important as the generator.

Chunking Affects Answer Completeness

Character-based chunking at 300 chars occasionally cut sentences mid-phrase, giving the LLM incomplete context. A sentence-aware chunker would improve this in future work.

Made with GAMMA

Conclusion and Future Work

What Was Achieved:

- Built a working RAG pipeline for UNH admissions Q&A
- Gemma-2b-it achieved avg score of 3.7/5 across 10 questions
- Demonstrated that instruction tuning is essential in RAG
- Identified and fixed retrieval failure cases through source-based boosting
- Compared RAG vs non-RAG baseline (DistilGPT2)
- Corpus of 7 official UNH admissions documents

Future Work

- Add FAISS vector index for faster retrieval at scale
- Implement sentence-aware chunking to avoid mid-phrase cuts
- Evaluate with ROUGE / BERTScore for automated metrics
- Expand corpus to cover program-specific pages
- Build a simple web UI for prospective students
- Write full technical report (ACL/NeurIPS format)

Made with GAMMA

Thank you!

Made with GAMMA